

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination- 2023 Course: B. Tech. Branch : All Semester :I Subject Code & Name: BTBS102P (Engineering Physics) Max Marks: 60 Date:09/08/2023 Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.			
A)	What is forced oscillations? Derive the differential equation for forced oscillations.	CO1	06	
B)	State piezoelectric effect and explain production of ultrasonic waves by using suitable diagram.	CO1	06	
C)	Write any two applications of ultrasonic in brief. Calculate the frequency of the piezoelectric crystal using the following data. (Vibrating length = 6 mm, Young's modulus = 8×10^{10} N/m ² and density of crystal = 2.5 g/cm ³)	CO1	06	
Q.2	Solve Any Two of the following.			
A)	Write a short note on wedge shaped film. Deduce the expression for fringe spacing in wedge shaped film.	CO2	06	
B)	Discuss the production of Laser light from Ruby laser with suitable diagrams.	CO2	06	
C)	Explain double refraction and distinguish between e-ray and o-ray. An optical fibre has a core material of refractive index of 1.55 and cladding material of refractive index of 1.50. The light is launched into it in air. Calculate the numerical aperture.	CO2	06	
Q. 3	Solve Any Two of the following.			
A)	Explain motion of electron in parallel and perpendicular electric field.	CO3	06	
B)	With the help of graph and suitable diagram discuss GM counter.	CO3	06	
C)	What is wave function? Derive Schrodinger's time independent wave equation.	CO3	06	
Q.4	Solve Any Two of the following.			
A)	What is packing density? Find the packing density for SC, BCC and FCC lattices.	CO4	06	
B)	Explain characteristics X-ray spectra with suitable diagrams. A Coolidge tube operates at 40 KV. Find the minimum wavelength of X rays generated.	CO4	06	
C)	Write four Maxwell's equations in integral form and write the physical significance of them.	CO4	06	
Q. 5	Solve the following.			
A)	On the basis of magnetic moment discuss diamagnetic, paramagnetic and ferromagnetic materials.	CO5	06	

B)	What is Meissner Effect? Prove that superconductors are act as a diamagnetic material using Meissner effect. The transition temperature for Pb is 7.2 K. However, at 5 K it losses the superconductivity property if subjected to a magnetic field of $3.3 \times 10^4 A/m$. Find the maximum value of H which allows the metal to retain its superconductivity at 0 K.	CO5	06
	*** End ***		